AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

ICICI Bank: AI-Driven Dashboard for Customer Financial Health & Risk Analysis

Project Domain

Excel / Power BI with Copilot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

This project is part of the Artificial Intelligence in the Contemporary World (AICW) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The project focuses on analyzing a dataset of 1600 ICICI Bank customer records to monitor financial health, identify high-risk and inactive accounts, and track loan exposure.

The problem addressed is the underutilization of banking data due to fragmentation, inconsistency, and lack of actionable insights. The objective is to develop an AI-assisted analytical solution that provides clear visualizations, automated insights, and actionable recommendations.

The methodology involves data cleaning and preprocessing using Microsoft Excel, analysis through pivot tables and formulas, and visualization using Power BI dashboards. AI integration is achieved using Microsoft Copilot and a custom-built AI agent, ICICI AI Assistant, developed using Copilot Studio. This AI agent automates insight generation, identifies high-risk customers, and assists in report creation.

Key results include a cleaned dataset, an interactive dashboard displaying account balances, loan exposure, and risk categories, and AI-generated insights to support decision-making. This project demonstrates the effectiveness of combining AI, business intelligence, and data analytics to enhance operational efficiency, risk management, and customer engagement in banking.

Keywords: AI in Banking, Data Analytics, Power BI, Microsoft Copilot, Customer Financial Health, Risk Assessment.

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CHAPTER 1

Introduction

This project is undertaken as part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, which focuses on building practical skills in data analytics, business intelligence, and AI-assisted reporting.

In today's digital banking environment, financial institutions generate vast amounts of customer data daily. However, much of this data remains underutilized due to the lack of centralized systems and analytical tools.This project aims to address this gap by transforming raw banking data into meaningful insights using Microsoft Excel, Power BI, and Microsoft Copilot.

The project focuses on analyzing customer financial behavior, identifying high-risk and inactive accounts, and monitoring loan exposure through AI-assisted insights. A key innovation of this project is the development of the ICICI AI Assistant using Microsoft Copilot Studio, which automates financial analysis and enhances decision-making efficiency.

1.1 Problem Statement

Describe the core problem your project addresses. Be specific and clear.

- **What is the main problem or challenge in your chosen domain?**

  The main problem in the banking domain is the underutilization of large volumes of customer data generated daily by banks such as ICICI Bank. This data is often scattered across multiple systems and lacks a centralized analytical framework. As a result, it becomes difficult for managers to extract meaningful insights related to customer behavior, financial health, and risk levels. Key challenges include identifying inactive accounts, detecting high-risk customers, monitoring loan exposure, and performing effective customer segmentation.

- **Why is this problem significant, and what is its real-world impact?**

  This problem is significant because improper utilization of data directly affects banking operations and decision-making. In the real world, it can lead to:

- Poor financial decisions due to lack of actionable insights
- Increased chances of loan defaults and financial losses
- Reduced customer engagement and retention
- Inefficient risk management
- If banks fail to analyze customer data effectively, they may not be able to identify risky customers or inactive accounts in time, which negatively impacts profitability and operational efficiency.

- **Who is affected by this problem — what are the target users or organizations?**

  The problem affects multiple stakeholders in the banking ecosystem, including:

- Banking managers and decision-makers
- Risk management teams
- Financial analysts
- Customer relationship managers

- Banking institutions such as ICICI Bank
- These users depend on accurate data insights to make strategic decisions, manage risks, and improve customer satisfaction.

- **What are the limitations of existing approaches or tools currently available?**

  The existing systems have several limitations, such as:
- Lack of a centralized dashboard for unified data analysis
- Limited ability to perform advanced customer segmentation
- Inability to detect unusual financial patterns effectively
- Absence of AI-driven insights and automation
- Time-consuming manual data analysis processes
- These limitations reduce efficiency and prevent banks from making fast and informed decisions.

- **Final Problem Statement (Conclusion)**

  Therefore, there is a need for an AI-driven analytical solution that can transform raw banking data into actionable insights. This project aims to develop a centralized dashboard using AI-powered tools to monitor customer financial health, identify inactive and high-risk accounts, track loan exposure, and support effective data-driven decision-making in the banking sector.

1.2 Motivation

Explain what motivated you to choose this particular project. Think about real-world relevance, personal interest, or professional value.

- **Why is this project personally or professionally meaningful to you?**

  This project is personally and professionally meaningful to me because it aligns with my strong interest in finance, investment banking, and data analytics. Working on an AI-driven dashboard helps me gain practical exposure to real-world tools like Power BI and enhances my understanding of how financial data is analyzed in banks. As someone aiming to build a career in the financial sector, this project helps me develop relevant skills such as data interpretation, risk analysis, and decision-making. It also bridges the gap between my academic knowledge and industry requirements.

- **What are the potential applications of this project in the real world?**

  This project has several real-world applications, especially in the banking sector. It can be used by financial institutions like ICICI Bank to:
- Identify high-risk customers and reduce chances of loan defaults
- Monitor customer financial health and spending patterns
- Segment customers for better marketing strategies
- Detect inactive accounts and improve customer engagement
- Support smarter loan approval and credit decisions
- Overall, the dashboard can help banks make faster, more accurate, and data-driven decisions.

- **What impact could this project have on the organization or domain?**

  This project can create a significant positive impact on the organization by improving efficiency, accuracy, and risk management. With better insights, bank managers can make informed decisions, reduce financial losses, and enhance customer satisfaction. It also helps in optimizing operations and improving overall performance. In the broader domain of banking and finance, such AI-based solutions promote innovation and encourage the adoption of smart technologies for better service delivery.

- **How does this project align with modern industry trends in AI and data analytics?**

  This project is highly aligned with current industry trends where organizations are increasingly using Artificial Intelligence (AI) and data analytics to drive business decisions. The banking sector is rapidly moving towards digital transformation, focusing on:

- AI-based risk assessment
- Predictive analytics for customer behavior
- Data visualization using tools like Power BI
- Automation of decision-making processes

  By developing this dashboard, the project reflects the growing importance of data-driven strategies and prepares for future advancements in AI-powered financial systems.

1.3 Objectives

Clearly state the objectives of your project. Objectives should be measurable and specific.

- To collect and understand the structure of the dataset used in the project, including key variables such as customer demographics, account details, transaction history, and loan-related information for effective analysis.

- To clean and preprocess raw data by removing duplicate entries, correcting inconsistent formats (such as date, currency, and numerical values), and handling missing or incomplete data to improve overall data quality and reliability.

- To apply domain-specific analysis using tools like Excel, Power BI, Microsoft Copilot and ICICI AI Assistant in order to identify patterns, trends, and relationships related to customer behavior, financial health, and risk factors.

- To build interactive dashboards and reports using advanced data visualization techniques that effectively present insights through charts, graphs, and key performance indicators (KPIs) for better understanding.

- To leverage Microsoft Copilot and the ICICI AI Assistant to automate repetitive tasks, enhance data analysis, generate meaningful insights, and assist in report creation.

- To develop an AI-driven dashboard for customer financial health and risk analysis in ICICI Bank, which can support better decision-making, improve risk management, and enhance overall customer relationship management.

1.4 Scope of the Project

Define the boundaries and limitations of the project clearly.

- This project covers data related to banking operations, including customer details, account information, transaction records, and loan-related data used for analysis.
- The tools used in this project are limited to Excel, Power BI, and Microsoft Copilot (ICICI AI Assistant) as assigned under the fellowship guidelines.
- The analysis mainly focuses on data cleaning, preprocessing, data visualization, dashboard creation, and generating insights related to customer financial health and risk analysis.
- The project emphasizes understanding customer behavior, identifying high-risk customers, and presenting insights through interactive dashboards for better decision-making.
- The scope is limited to historical and static data analysis and does not include real-time data integration or live system updates.
- The project does not involve advanced predictive machine learning models or complex AI model development, and is focused primarily on descriptive and analytical insights.
- The project is developed for academic and learning purposes as part of a fellowship showcase and is not intended to be a fully functional production-level system.
- The analysis and results are based only on the available dataset and may vary if different or larger datasets are used in real-world scenarios.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

In the domain of banking and financial analytics, multiple tools and techniques are used to process and analyze large datasets. Microsoft Excel has been widely applied for data cleaning, preprocessing, and basic visualization, including removing duplicates, handling missing values, applying conditional formatting, and using pivot tables to summarize customer transactions and account details. While Excel provides basic insights, it is limited when analyzing large datasets with complex structures.

Microsoft Power BI has been widely adopted for interactive dashboard creation and business intelligence reporting. It allows integration of multiple datasets, creation of dynamic reports, and monitoring of key performance indicators (KPIs) such as customer financial behavior, loan exposure, and transaction trends. Several Power BI templates exist, but most focus on general business metrics rather than domain-specific banking analysis.

Recent advancements in AI-assisted tools like Microsoft Copilot and AI-assisted tools such as Microsoft Copilot enable automation of repetitive tasks, prompt-driven analysis, and generation of insights. Copilot assists in summarizing datasets, performing calculations, and generating reports using natural language, while AI chatbots provide an interactive approach for interpreting data and generating business-focused content.

Despite these developments, gaps remain:

- Existing solutions either rely on manual Excel analysis or focus on advanced AI/ML models, which can be inaccessible for non-technical users.
- Limited research exists on integrating AI Copilot with Excel and Power BI for banking-specific analytics.
- Prompt engineering best practices for financial data analysis are evolving and not standardized.
- Dashboards rarely combine data cleaning, visualization, and AI-generated insights in one workflow, reducing efficiency.

How this project addresses the gaps:

This project creates an integrated solution using Excel, Power BI, and AI tools for customer financial health and risk analysis in ICICI Bank. By combining data cleaning, interactive visualization, and AI-assisted insights, it simplifies analysis for non-technical users and provides actionable insights for decision-making, risk assessment, and customer management.

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2.2 Tools and Technologies Review

Briefly review the tools and technologies used in your project:

- **Microsoft Excel** – Excel is a widely used tool for data cleaning, preprocessing, and basic visualization [4]. In this project, Excel was employed to remove duplicate records, handle missing values, correct inconsistent formats, and organize banking data such as customer demographics, account details, and transaction histories. Advanced features like pivot tables, conditional formatting, and data validation were leveraged to structure and summarize the dataset effectively before further analysis.

- **Microsoft Power BI** – Power BI is a business analytics and data visualization tool that allows creation of interactive dashboards and business intelligence reports [1]. In this project, Power BI was used to integrate multiple datasets, create dynamic visualizations, and monitor key financial metrics and customer risk patterns. This enables ICICI Bank stakeholders to make informed decisions by presenting complex data in a simple and interactive format.

- **Microsoft Copilot & ICICI AI Assistant** – Microsoft Copilot is an AI-powered assistant integrated into Microsoft 365 tools that helps automate tasks, generate insights, and perform analyses using natural language prompts [6]. For this project, an AI agent named ICICI AI Assistant was created using Copilot Studio. The agent automates analysis of customer financial data, identifies high-risk customers, generates actionable insights, and assists in report creation. By using ICICI AI Assistant, repetitive tasks are reduced, analysis becomes faster, and decision-making is enhanced for banking operations.

- **Power Query (Excel add-in)** – Power Query was used for advanced data transformation, merging multiple datasets, and cleaning complex banking data before integrating it into Power BI dashboards. This tool ensures accurate and efficient data preprocessing.

- **Other Tools** – Standard office tools like Microsoft Word and PowerPoint were used for documenting findings, preparing reports, and presenting dashboards to stakeholders.

Summary:

The combination of Excel, Power BI, and Microsoft Copilot (through ICICI AI Assistant) provides a comprehensive workflow for the project, covering data cleaning, analysis, visualization, and AI-assisted insight generation. This integration ensures that the AI-driven dashboard is efficient, interactive, and actionable, tailored specifically for ICICI Bank's customer financial health and risk assessment.

2.3 Gaps Identified

Identify gaps in existing literature or tools relevant to your project domain.

- Most existing literature and solutions focus on complex machine learning or AI algorithms for financial risk analysis, which are often difficult for non-technical users to implement. There is a lack of simpler, tool-based solutions that combine usability with actionable insights.

- There is limited documentation on integrating Microsoft Copilot with Excel and Power BI for domain-specific banking analysis, particularly for use in student projects or practical dashboard development. Most tutorials focus on generic Copilot applications rather than financial dashboards.

- While prompt engineering is becoming increasingly popular for AI tools, best practices for business and banking use cases are still evolving. There is no standardized framework for creating AI agents like ICICI AI Assistant that can automate analysis, generate insights, and assist decision-making within a banking context.

- Existing dashboards and financial analytics tools rarely combine data cleaning, interactive visualization, and AI-assisted insights into a single workflow. This creates inefficiencies and makes it difficult for decision-makers to access actionable information quickly.

- Domain-specific application gaps: In the context of Indian banks like ICICI, there are very few custom AI solutions designed to analyze customer financial health and risk patterns in an integrated, user-friendly dashboard. Most studies focus on general finance analytics, missing the specific challenges of banking datasets and risk assessment.

- **How this project addresses the gaps:**

- This project develops an AI-driven dashboard using Excel, Power BI, and ICICI AI Assistant (built on Microsoft Copilot). It addresses the above gaps by providing:
- A simpler, tool-based solution accessible to non-technical users.
- Seamless integration of Copilot with Excel and Power BI for domain-specific banking analysis.
- A custom AI agent that automates repetitive tasks, generates insights, and assists in report creation.
- A unified workflow that combines data cleaning, visualization, and AI-assisted insights for actionable decision-making.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The system is designed to transform raw banking data into meaningful insights through a structured process involving data cleaning, analysis, visualization, and AI integration. The overall workflow ensures efficient data processing and supports data-driven decision-making.

- **Data Source / Input:**

  The input data consists of a simulated dataset containing 1,600 ICICI Bank customer records. The dataset includes details such as customer name, account type, account balance, transaction count, loan amount, loan status, customer segment, and risk category. The data is stored in Excel format and serves as the foundation for analysis.
- **Data Cleaning:**

  The dataset is cleaned using Microsoft Excel to ensure accuracy and consistency. This includes removing duplicate records, handling missing values, correcting data types, and standardizing formats. Functions such as IF, IFERROR, and Text-to-Columns are used to improve data quality.
- **Analysis / Processing:**

  The cleaned data is analyzed using Excel formulas, pivot tables, and Power BI measures. Key financial indicators such as total loan amount, average balance, transaction count, and risk distribution are calculated. AI tools like Microsoft Copilot and ChatGPT are used to generate insights, identify patterns, and provide recommendations.
- **Visualization / Output:**

  The final output is an interactive dashboard created in Power BI. The dashboard includes KPI cards, charts, and visualizations that display customer segmentation, loan exposure, account status (active/inactive), and risk categories. This enables users to quickly interpret data and supports efficient data-driven decisions-making.

3.2 Modules / Components Used

The system is divided into the following functional modules:

- **Module 1 — Data Ingestion:**

  This module involves loading the raw dataset into Microsoft Excel and Power BI for further processing.
- **Module 2 — Data Cleaning:**

  In this stage, inconsistencies such as duplicate entries, missing values, and incorrect formats are identified and corrected to ensure data accuracy.
- **Module 3 — Data Analysis:**

  This module includes applying Excel formulas, pivot tables, and Power BI DAX measures to analyze customer data, calculate KPIs, and identify trends and patterns.

- **Module 4 — Visualization / Report Generation:**
  The processed data is visualized using Power BI dashboards. Charts, graphs, and KPI cards are created to present insights in an easy-to-understand format.
- **Module 5 — AI Integration:**
  AI tools such as Microsoft Copilot and ChatGPT are used to generate automated insights, summaries, and recommendations, enhancing the analytical capabilities of the system.
- **Module 6 — Insight Generation & Decision Support:**
  This module focuses on converting analyzed data into actionable insights. It helps identify inactive accounts, high-risk customers, and loan exposure patterns, supporting better decision-making and strategic planning.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows through a system. It shows the movement of data between inputs, processes, and outputs, helping to understand the overall system structure and workflow. In this project, two levels of DFD are used to represent the system: Level 0 (Context Diagram) and Level 1 (Detailed Process Flow).

DFD Level 0 — Context Diagram

The Level 0 DFD provides a high-level overview of the system, representing it as a single process. It shows how raw data is transformed into meaningful outputs through the dashboard system.

In this project, the input consists of raw customer data from the ICICI Bank dataset. This data is processed within the AI-driven dashboard system, which performs cleaning, analysis, and visualization. The output generated includes cleaned data, interactive dashboards, and actionable insights

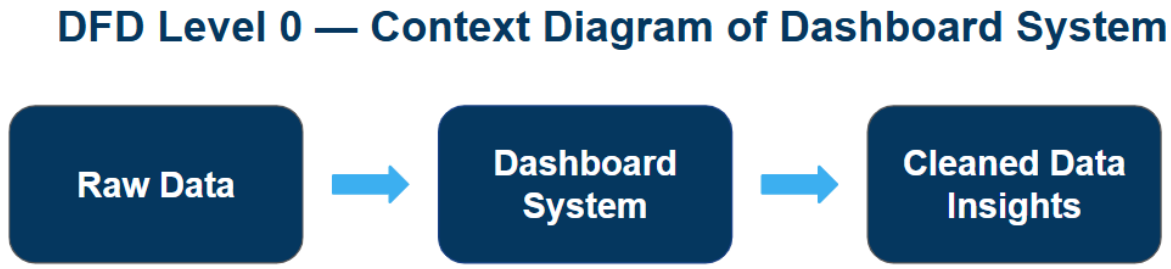


Figure 6: DFD Level 0 — Context Diagram of Dashboard System.

DFD Level 1 — Detailed Process Flow

The Level 1 DFD provides a detailed breakdown of the internal processes within the system. It explains how data moves through multiple stages to produce the final output.

The system is divided into the following sub-processes:

- **Data Loading and Validation:** The raw dataset is imported into Excel/Power BI and checked for missing values, duplicates, and incorrect formats.
- **Data Cleaning and Transformation:** Data is cleaned by removing duplicates, handling null values, and converting columns into appropriate formats for analysis.
- **Analysis and AI Processing:** Financial metrics such as loan amount, balances, and risk categories are analyzed using Excel formulas, Power BI measures, and AI tools like Microsoft Copilot to generate insights.
- **Visualization and Report Generation:** The processed data is visualized using Power BI dashboards, including charts, KPI cards, and summaries for easy interpretation.

The final output includes a centralized dashboard along with AI-generated insights for decision-making.

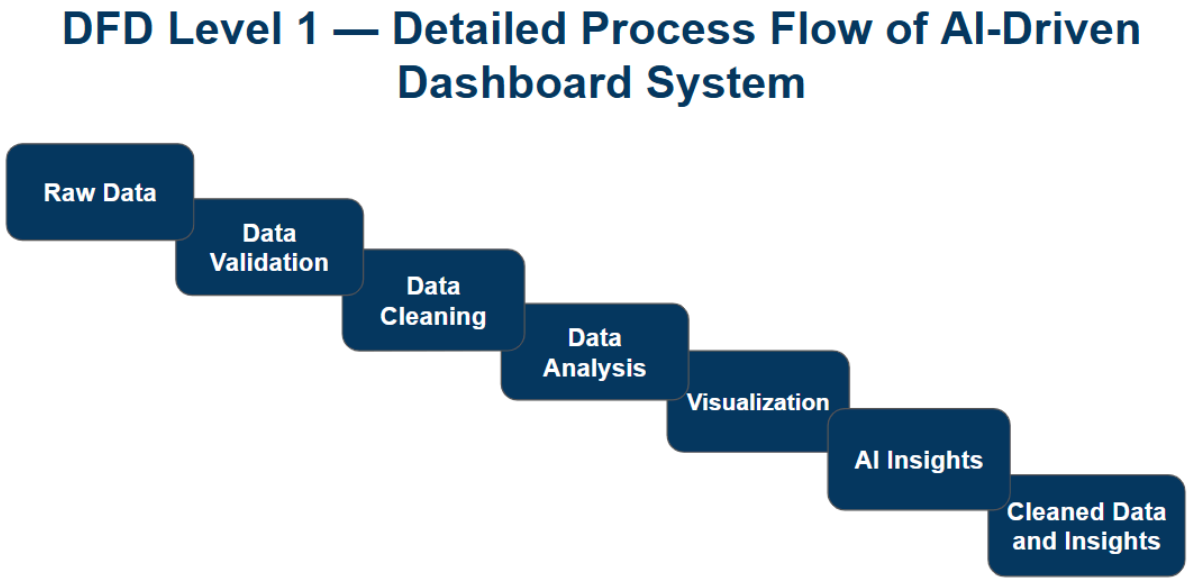


Figure 7: DFD Level 1— Detailed Process Flow of AI-Driven Dashboard System.

3.4 Advantages

The proposed AI-driven dashboard system offers several advantages that enhance efficiency, accuracy, and decision-making in banking analytics:

- **Automation of Data Processing:**
  The system automates repetitive data cleaning and preprocessing tasks, significantly reducing manual effort and saving time.
- **Improved Data Accuracy:**
  By standardizing formats and removing inconsistencies, the system ensures reliable and error-free data for analysis.
- **Interactive Visualization:**
  The use of Power BI dashboards provides clear, dynamic, and visually appealing insights that are easy to understand, even for non-technical users.
- **AI-Assisted Insights:**
  Integration of AI tools such as Microsoft Copilot and ChatGPT enables faster analysis, automated summaries, and intelligent recommendations, reducing human error.
- **Enhanced Decision-Making:**
  The dashboard provides actionable insights on customer behavior, risk levels, and loan exposure, supporting data-driven strategic decisions.
- **Scalability:**
  The methodology can be easily applied to larger datasets or extended to other domains such as finance, healthcare, or marketing analytics.
- **Cost-Effective Solution:**
  The project uses widely available tools like Microsoft Excel and Power BI, making it economical and accessible.
- **Improved Risk Management:**
  Helps in identifying high-risk customers and inactive accounts, enabling proactive measures for risk mitigation and customer engagement.
- **Centralized Data Analysis:**
  Combines multiple data aspects into a single dashboard, eliminating fragmentation and improving efficiency.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Data cleaning is a crucial step in any data analytics project because raw datasets often contain inconsistencies, formatting issues, or duplicate values that can affect the accuracy of analysis. Before performing any analysis, the dataset was carefully inspected and cleaned using Microsoft Excel.

- **Dataset Source:**

  The dataset used in this project is a manually created synthetic banking customer dataset, developed for analytical and educational purposes as part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program.

  It simulates real-world banking customer records and includes information such as customer ID, customer segment, account type, account balances, loan details, transaction activity, account status, and risk category. Creating the dataset manually allowed demonstration of data cleaning, analysis, and visualization in a structured environment.

- **Original Dataset Size:**

  Rows: 1,600

  Columns: 16

- **Fields in the Raw Dataset:**

  Customer ID, Customer Name, Customer Segment, Account Type, Branch Region, Account Open Date, Opening Balance, Total Credits, Total Debits, Average Balance, Current Balance, Loan Amount, Loan Status, Account Status, Risk Category, Transaction Count.

- **Issues Found in the Raw Dataset:**

- Extra spaces in text fields such as Customer Segment and Account Type
- Minor formatting inconsistencies in categorical values
- Date formatting inconsistencies in the Account Open Date column
- Possible duplicate entries requiring verification
- Numeric fields requiring validation to ensure correct number formatting

- **Cleaning Steps Applied:**

  The dataset was cleaned using Microsoft Excel tools and functions. The steps performed include:

- **Duplicate Verification** – Checked Customer ID for uniqueness; no duplicates were present.
- **Text Cleaning** – Used TRIM() and CLEAN() functions to remove extra spaces and formatting errors.
- **Date Standardization** – Converted Account Open Date into a consistent date format.
- Numeric Data Validation – Verified columns like Opening Balance, Total Credits, Total Debits, Loan Amount, and Transaction Count contained valid numeric values.
- **Categorical Data Standardization** – Standardized Loan Status, Account Status, and Risk Category to maintain consistent labels.

- **Cleaned Dataset Size:**

  Rows: 1,600

  Columns: 16

No records were removed, and the dataset remained complete and ready for analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Customer	Customer_Name (St	Customer	Account_	Branch_R	Account_	Opening_	Total_Cre	Total_Del	Average_	Current_E	Loan_Am	Loan_Stat	Account_	Risk_Cate	Transactio
2	CUST0001	Vivaan Kapoor	SME	Current	West	22/08/202	179093.1	294928	203891.6	224611.3	270129.5	152681.4	Closed	Inactive	Medium	176
3	CUST0002	Ishita Mehta	SME	Savings	North	02/01/202	191192.5	156187	121939	208316.5	225440.5	238716.9	NPA	Inactive	Medium	17
4	CUST0003	Manish Yadav	SME	Savings	South	06/09/202	42866.88	401588.3	50395.26	218463.4	394059.9	253163.5	NPA	Inactive	Low	80
5	CUST0004	Akash Malhotra	Corporate	Savings	North	19/04/201	47920.74	324981.5	248119.2	86351.91	124783.1	157731	NPA	Inactive	Low	132
6	CUST0005	Manish Gupta	Retail	Business	East	01/03/202	161982.6	163289.7	264376	111439.5	60896.29	118309	NPA	Inactive	Low	177
7	CUST0006	Akash Reddy	SME	Business	North	22/01/202	37137.13	137605.4	153313.9	29282.88	21428.63	60852.51	Closed	Active	Medium	117
8	CUST0007	Priya Verma	SME	Savings	West	18/02/201	94376.26	157636.9	99379.5	123505	152633.7	248598.9	NPA	Inactive	High	104
9	CUST0008	Rahul Das	SME	Salary	North	29/12/202	51015.64	35846.18	393774.3	-127948	-306912	57853.99	Closed	Active	High	6
10	CUST0009	Aman Bansal	Corporate	Current	East	06/07/201	190146.1	32045.27	207674	102331.7	14517.34	25086.95	No Loan	Inactive	Medium	133
11	CUST0010	Suresh Nair	SME	Salary	North	06/02/202	183912.8	79103.12	95043.84	175942.5	167972.1	80206.54	Active	Active	High	187
12	CUST0011	Shreya Verma	Retail	Savings	South	04/10/201	75111.16	157910.3	50818.52	128657.1	182202.9	212791.5	Active	Inactive	High	161
13	CUST0012	Varun Reddy	SME	Salary	South	06/11/201	167089.5	421129.4	362321.9	196493.3	225897	114500.9	NPA	Inactive	Low	35
14	CUST0013	Varun Nair	Retail	Salary	West	13/08/202	9944.47	167633	290385	-51431.5	-112807	40058.92	No Loan	Active	Low	126
15	CUST0014	Aditi Singh	Corporate	Current	East	21/04/201	89656.24	163464	185148.2	78814.11	67971.97	215329.9	No Loan	Active	Medium	38
16	CUST0015	Simran Kapoor	Corporate	Business	North	22/11/202	190382.2	418267.7	354817.9	222107.2	253832.1	168545.7	Active	Inactive	Medium	200
17	CUST0016	Varun Reddy	SME	Business	South	24/07/201	163440.7	145128.1	208495.2	131757.2	100073.6	243471.5	Closed	Inactive	High	192
18	CUST0017	Diya Gupta	Corporate	Salary	South	05/04/201	107775	41363.5	363377.5	-53231.9	-214239	144401.2	Active	Inactive	Low	84
19	CUST0018	Manish Patel	SME	Business	North	20/10/201	140199.8	306666.6	106390	240338.1	340476.4	236600.1	No Loan	Active	Medium	66
20	CUST0019	Shreya Verma	SME	Business	North	26/07/202	23465.74	376230	299229.2	61966.15	100466.6	240555.4	Active	Active	Medium	136
21	CUST0020	Simran Patel	Retail	Salary	South	23/04/202	128273.7	14166.66	154935.2	57889.41	-12494.9	233292.4	No Loan	Active	Low	142
22	CUST0021	Suresh Khan	SME	Business	North	27/08/202	42752.07	222365.9	346412.4	-19271.2	-81294.4	181084.2	Active	Inactive	Low	182
23	CUST0022	Vihaan Verma	Corporate	Salary	North	22/08/201	104670.2	52833.85	94987.36	83593.48	62516.72	227026.5	Closed	Active	Low	191
24	CUST0023	Aditya Nair	Retail	Salary	South	11/07/201	12998.25	200466.8	274989.5	-24263.1	-61524.4	134643.6	NPA	Inactive	Low	154
25	CUST0024	Vivaan Reddy	SME	Current	West	08/12/202	158501.5	452852.3	204048.7	282903.2	407305	84630.13	No Loan	Active	Medium	127
26	CUST0025	Vihaan Joshi	Retail	Salary	West	25/05/202	158149.9	343103.2	14331.65	322535.6	486921.4	79248.27	NPA	Inactive	Low	56
27	CUST0026	Ananya Singh	SME	Salary	North	04/05/201	98931.68	11434.92	137296.2	36001.05	-26929.6	164076.7	Active	Inactive	Low	41
28	CUST0027	Diya Mehta	Retail	Salary	West	13/11/201	5938.22	174323.2	306492.6	-60146.5	-126231	140314.3	No Loan	Active	High	24
29	CUST0028	Shreya Yadav	Retail	Current	South	14/07/201	31689.4	20207.69	270643	-93528.2	-218746	117231.5	Active	Inactive	Low	42
30	CUST0029	Aditya Mehta	Retail	Salary	South	07/08/202	97849.51	93238.51	174261.4	57338.09	16826.67	201673.1	Active	Inactive	Low	59
31	CUST0030	Nikhil Reddy	SME	Business	South	06/01/201	188116.6	124200.4	105176	197628.9	207141.1	152414.1	Active	Inactive	Low	90
32	CUST0031	Sneha Khan	Corporate	Salary	North	20/01/201	91311.67	323438	275891.8	115084.8	138857.9	6326.96	Closed	Active	Low	100
33	CUST0032	Shreya Kapoor	SME	Current	West	08/09/201	71851.31	263539	353416.2	26912.74	-18025.8	139478.9	No Loan	Inactive	Low	74
34	CUST0033	Sai Patel	SME	Salary	West	08/08/202	112692.6	32119.65	61165.15	98169.83	83647.08	225323.2	No Loan	Inactive	Medium	80
35	CUST0034	Sneha Das	Retail	Savings	East	15/11/202	125413.7	91774	349644.7	-3521.67	-132457	58853.62	Active	Active	Medium	65
36	CUST0035	Kavya Joshi	Corporate	Salary	South	09/08/202	111164.4	302461	283691.5	120549.2	129933.9	3181.56	No Loan	Active	Low	39
37	CUST0036	Meera Malhotra	SME	Current	West	30/07/202	52749.75	22503.82	39716.76	44143.28	35536.81	107159	No Loan	Inactive	High	12
38	CUST0037	Meera Verma	SME	Savings	North	02/10/201	112736.5	51104.45	277329.5	-376.03	-113489	32272.75	No Loan	Inactive	Medium	108
39	CUST0038	Kavya Kapoor	Retail	Current	South	13/09/201	63631.09	497523	166969.5	228907.8	394184.6	25591.76	No Loan	Active	High	97
40	CUST0039	Vivaan Khan	SME	Current	South	22/07/202	120151.3	66402.69	75612.7	115546.3	110941.3	10555.27	Closed	Active	Low	7

Insert Figure 1: Screenshot of the raw (uncleaned) dataset.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
ICICI Bank: AI-Driven Dashboard for Customer Financial Health & Risk Analysis															
Customer ID	Customer Name	Customer Segment	Account Type	Branch Region	Account Open Date (DD/MM/YYYY)	Opening Balance	Total Credits	Total Debits	Average Balance	Current Balance	Loan Amount	Loan Status	Account Status	Risk Category	Transaction Count
CUST0001	Vivaan Kapoor	SME	Current	West	22/08/2021	179093.1	294928.02	203891.6	224611.31	270129.52	152681.44	Closed	Inactive	Medium	176
CUST0002	Ishita Mehta	SME	Savings	North	02/01/2025	191192.45	156187.03	121939	208316.47	225440.48	238716.94	NPA	Inactive	Medium	17
CUST0003	Manish Yadav	SME	Savings	South	06/09/2021	42866.88	401588.31	50395.26	218463.4	394059.93	253163.52	NPA	Inactive	Low	80
CUST0004	Akash Malhotra	Corporate	Savings	North	19/04/2019	47920.74	324981.53	248119.19	86351.91	124783.08	157731.03	NPA	Inactive	Low	132
CUST0005	Manish Gupta	Retail	Business	East	01/03/2020	161982.62	163289.68	264376.01	111439.46	60896.29	118309.01	NPA	Inactive	Low	177
CUST0006	Akash Reddy	SME	Business	North	22/01/2024	37137.13	137605.37	153313.87	29282.88	21428.63	60852.51	Closed	Active	Medium	117
CUST0007	Priya Verma	SME	Savings	West	18/02/2015	94376.26	157636.91	99379.5	123504.96	152633.67	248598.89	NPA	Inactive	High	104
CUST0008	Rahul Das	SME	Salary	North	29/12/2022	51015.64	35846.18	393774.26	-127948.4	-306912.44	57853.99	Closed	Active	High	6
CUST0009	Aman Bansal	Corporate	Current	East	06/07/2016	190146.05	32045.27	207673.98	102331.69	14517.34	25086.95	No Loan	Inactive	Medium	133
CUST0010	Suresh Nair	SME	Salary	North	06/02/2022	183912.82	79103.12	95043.84	175942.46	167972.1	80206.54	Active	Active	High	187
CUST0011	Shreya Verma	Retail	Savings	South	04/10/2019	75111.16	157910.3	50818.52	128657.05	182202.94	212791.49	Active	Inactive	High	161
CUST0012	Varun Reddy	SME	Salary	South	06/11/2019	167089.53	421129.44	362321.93	196493.29	225897.04	114500.87	NPA	Inactive	Low	35
CUST0013	Varun Nair	Retail	Salary	West	13/08/2022	9944.47	167633.04	290384.95	-51431.49	-112807.44	40058.92	No Loan	Active	Low	126
CUST0014	Aditi Singh	Corporate	Current	East	21/04/2019	89656.24	163463.97	185148.24	78814.11	67971.97	215329.94	No Loan	Active	Medium	38
CUST0015	Simran Kapoor	Corporate	Business	North	22/11/2022	190382.22	418267.72	354817.85	222107.15	253832.09	168545.66	Active	Inactive	Medium	200
CUST0016	Varun Reddy	SME	Business	South	24/07/2017	163440.72	145128.06	208495.15	131757.17	100073.63	243471.47	Closed	Inactive	High	192
CUST0017	Diya Gupta	Corporate	Salary	South	05/04/2016	107775.04	41363.5	363377.45	-53231.94	-214238.91	144401.18	Active	Inactive	Low	84
CUST0018	Manish Patel	SME	Business	North	20/10/2016	140199.83	306666.61	106390.01	240338.13	340476.43	236600.12	No Loan	Active	Medium	66
CUST0019	Shreya Verma	SME	Business	North	26/07/2021	23465.74	376230.01	299229.2	61966.15	100466.55	240555.43	Active	Active	Medium	136
CUST0020	Simran Patel	Retail	Salary	South	23/04/2020	128273.68	14166.66	154935.2	57889.41	-12494.86	233292.42	No Loan	Active	Low	142
CUST0021	Suresh Khan	SME	Business	North	27/08/2025	42752.07	222365.9	346412.39	-19271.17	-81294.42	181084.15	Active	Inactive	Low	182
CUST0022	Vihaan Verma	Corporate	Salary	North	22/08/2016	104670.23	52833.85	94987.36	83593.48	62516.72	227026.5	Closed	Active	Low	191
CUST0023	Aditya Nair	Retail	Salary	South	11/07/2018	12998.25	200466.83	274989.49	-24263.08	-61524.41	134643.61	NPA	Inactive	Low	154
CUST0024	Vivaan Reddy	SME	Current	West	08/12/2023	158501.45	452852.25	204048.69	282903.23	407305.01	84630.13	No Loan	Active	Medium	127
CUST0025	Vihaan Joshi	Retail	Salary	West	25/05/2020	158149.86	343103.21	14331.65	322535.64	486921.42	79248.27	NPA	Inactive	Low	56
CUST0026	Ananya Singh	SME	Salary	North	04/05/2016	98931.68	11434.92	137296.17	36001.05	-26929.57	164076.7	Active	Inactive	Low	41
CUST0027	Diya Mehta	Retail	Salary	West	13/11/2017	5938.22	174323.24	306492.6	-60146.46	-126231.14	140314.28	No Loan	Active	High	24
CUST0028	Shreya Yadav	Retail	Current	South	14/07/2019	31689.4	20207.69	270642.96	-93528.24	-218745.87	117231.49	Active	Inactive	Low	42
CUST0029	Aditya Mehta	Retail	Salary	South	07/08/2023	97849.51	93238.51	174261.35	57338.09	16826.67	201673.08	Active	Inactive	Low	59
CUST0030	Nikhil Reddy	SME	Business	South	06/01/2017	188116.64	124200.4	105175.97	197628.86	207141.07	152414.11	Active	Inactive	Low	90
CUST0031	Sneha Khan	Corporate	Salary	North	20/01/2019	91311.67	323438	275891.75	115084.8	138857.92	6326.96	Closed	Active	Low	100
CUST0032	Shreya Kapoor	SME	Current	West	08/09/2019	71851.31	263539	353416.15	26912.74	-18025.84	139478.87	No Loan	Inactive	Low	74
CUST0033	Sai Patel	SME	Salary	West	08/08/2021	112692.58	32119.65	61165.15	98169.83	83647.08	225323.2	No Loan	Inactive	Medium	80
CUST0034	Sneha Das	Retail	Savings	East	15/11/2021	125413.68	91774	349644.7	-3521.67	-132457.02	58853.62	Active	Active	Medium	65
CUST0035	Kavya Joshi	Corporate	Salary	South	09/08/2022	111164.37	302461.01	283691.45	120549.15	129933.93	3181.56	No Loan	Active	Low	39
CUST0036	Meera Malhotra	SME	Current	West	30/07/2024	52749.75	22503.82	39716.76	44143.28	35536.81	107158.99	No Loan	Inactive	High	12
CUST0037	Meera Verma	SME	Savings	North	02/10/2015	112736.47	51104.45	277329.45	-376.03	-113488.53	32272.75	No Loan	Inactive	Medium	108
CUST0038	Kavya Kapoor	Retail	Current	South	13/09/2015	63631.09	497523	166969.53	228907.83	394184.56	25591.76	No Loan	Active	High	97
CUST0039	Vivaan Khan	SME	Current	South	22/07/2021	120151.3	66402.69	75612.7	115546.29	110941.29	10555.27	Closed	Active	Low	7
CUST0040	Aarav Patel	Corporate	Business	North	17/03/2016	92486.99	411218.25	292661.34	151765.45	211043.9	249421.18	No Loan	Active	Medium	151
CUST0041	Aditya Mehta	Corporate	Business	North	18/04/2020	48788.74	193741.27	298945.13	-3813.19	-56415.12	83604.93	Active	Inactive	Medium	142
CUST0042	Reyansh Joshi	SME	Current	North	21/08/2025	93455.92	291795.87	120228.5	179239.6	265023.29	98214.37	NPA	Active	Low	60
CUST0043	Neha Patel	SME	Business	North	19/06/2024	29298.39	195095.24	11380.58	121155.72	213013.05	51994.13	NPA	Active	High	142
CUST0044	Sneha Khan	SME	Current	South	04/04/2017	116835.8	428857.9	299291.92	181618.79	246401.78	133493.14	No Loan	Inactive	Medium	74
CUST0045	Tanya Nair	Corporate	Savings	South	10/03/2019	10429.91	335677.46	144149.34	106193.97	201958.03	127560.5	Active	Active	Low	188
CUST0046	Reyansh Choudhary	Corporate	Savings	North	11/09/2025	98217.43	170386.26	208158.7	79331.21	60444.99	283914.25	Active	Inactive	High	174

Insert Figure 2: Screenshot of the cleaned ICICI Bank dataset after preprocessing.

Description in the report:

"The cleaned dataset, shown in Figure 2, highlights the preprocessed banking data, including customer demographics, account details, transaction history, and loan information. All duplicate records have been removed, missing values have been handled, and inconsistent formats have been corrected. This preprocessing ensures that the dataset is accurate, reliable, and ready for further analysis using Power BI and the ICICI AI Assistant."

4.2 Visualization / Output Results

This section presents the key visualizations and dashboards created using Power BI to analyze the dataset effectively. These visuals help in understanding customer behavior, financial trends, and overall banking performance.

- **Dashboard / Chart 1: Customer Segmentation Analysis (Pie Chart)**

  This visualization shows the distribution of customers across different segments such as Corporate, SME, and Retail.
- **Insight:**

  The chart highlights a balanced customer distribution, which supports diversified business operations and reduces dependency on a single segment.

- **Dashboard / Chart 2: Account Activity Status (Bar Chart)**

  This chart represents the number of active and inactive accounts in the dataset.

- **Insight:**
  The visualization provides a clear view of customer engagement levels and helps in understanding overall account activity patterns.

- **Dashboard / Chart 3: Loan Status Distribution (Donut Chart)**
  This chart displays different loan categories such as Active Loans, Closed Loans, No Loans, and NPA.
- **Insight:**
  It reflects the loan portfolio structure and shows customer participation in banking credit services.

- **Dashboard / Chart 4: Risk Category Analysis (Column Chart)**
  This visualization categorizes customers into Low Risk, Medium Risk, and High Risk segments.
- **Insight:**
  It provides a clear overview of risk distribution, supporting strategic planning and decision-making.

- **Dashboard / Chart 5: Financial Performance Overview (KPI Cards)**
  This includes key metrics such as:
- Average Balance
- Total Credits
- Total Debits
- Loan Amount

- **Insight:**
- These KPIs offer a quick snapshot of overall financial performance and customer activity.

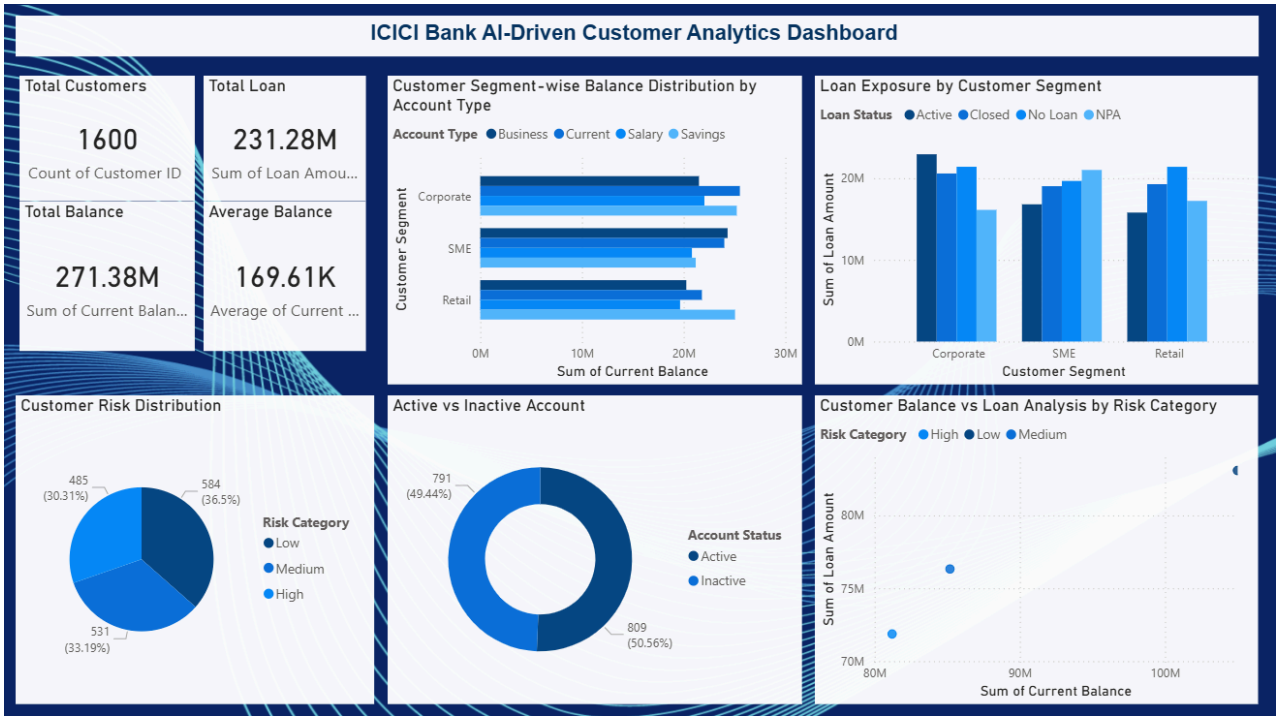
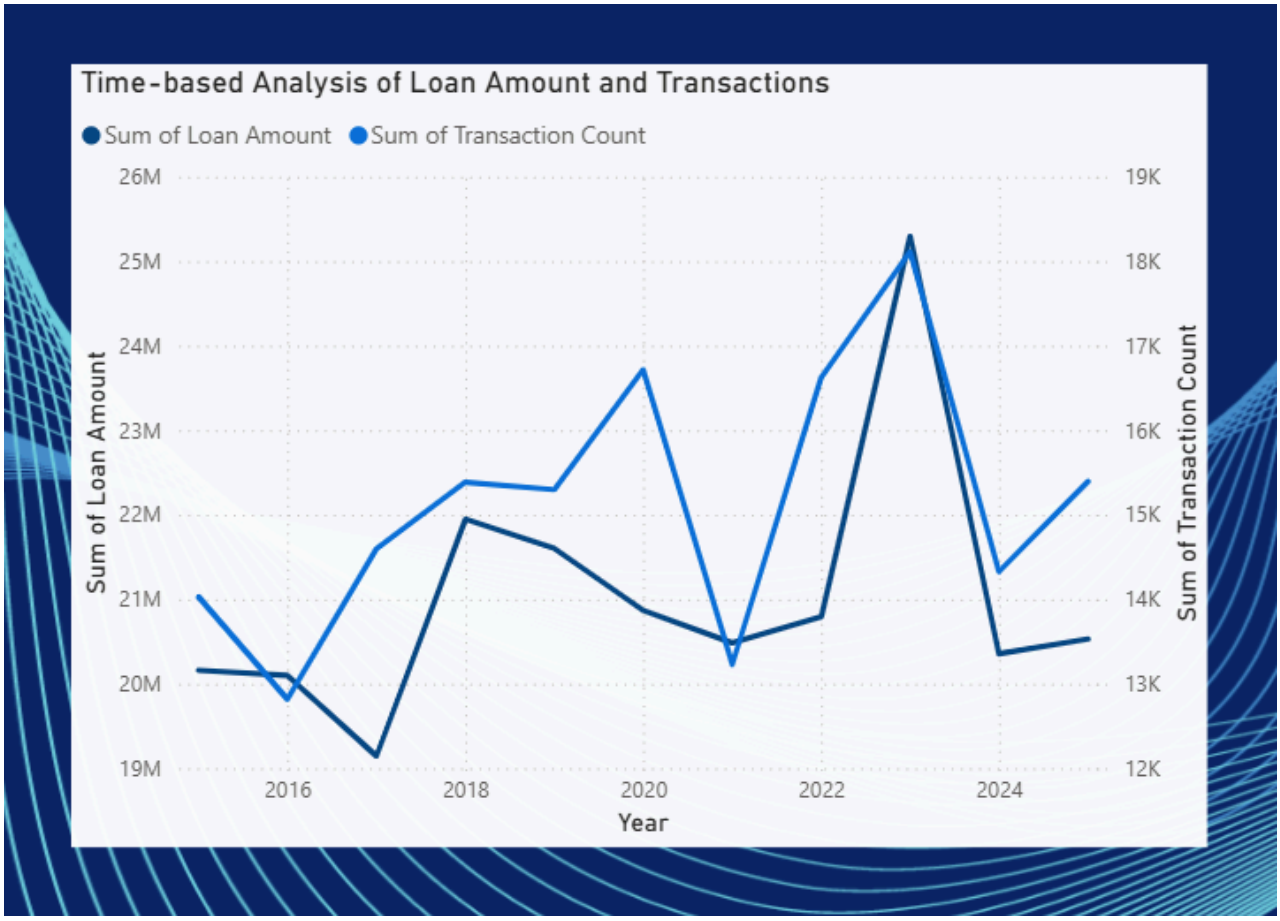


Figure 3: Main Power BI Dashboard showing customer segmentation, account activity, loan distribution, risk analysis, and key financial KPIs

The above dashboard provides a comprehensive overview of the banking dataset through interactive visualizations. It includes key insights such as customer segmentation, account activity status, loan portfolio distribution, and risk categorization. Additionally, KPI cards display important financial metrics like total balance, credits, and loan amounts. The dashboard enables easy interpretation of data and supports effective decision-making.



Insert Figure 4: Time-based Analysis of Loan Amount and Transactions.

4.3 AI Tool Integration Results

- **Tool Used: Microsoft Copilot**

- **Prompts Used:**
- "Analyze ICICI Bank customer data to identify high-risk customers and loan exposure patterns."
- "Summarize inactive accounts and their impact on overall banking operations."
- "Provide insights on customer segmentation, including average balances, transaction activity, and risk categories."

- **Output Quality:**
- The AI-generated output was clear, structured, and actionable. Microsoft Copilot provided:
- Identification of high-risk customers, including NPA accounts, inactive accounts, and unusual balances.
- Loan exposure patterns across segments, highlighting SME and Corporate portfolio vulnerabilities.
- Segment-level insights showing customer risk trends, loan exposure, and transactional behavior.
- Strategic recommendations for early warnings, risk control, and customer engagement.

- **How it Enhanced the Project:**
- Faster Analysis: Automated the identification of high-risk accounts and loan exposure patterns, reducing manual effort.
- Better Insights: Generated structured insights and recommendations that were easy to interpret and act upon.
- Data-Driven Decisions: Helped highlight trends and anomalies that support proactive financial and operational decisions.
- Enhanced Accuracy: Reduced human error in risk assessment and segmentation analysis.

- **Example Insights Produced:**
- Customers with negative balances or irregular transactions flagged as high risk.
- High-risk NPAs concentrated in SME and Retail segments.
- Inactive accounts with outstanding loans identified for potential consolidation or reactivation.
- Tailored recommendations for risk mitigation and customer engagement.

Prompts Used:
"Analyze ICICI Bank customer data to identify high-risk customers and loan exposure patterns."
"Summarize inactive accounts and their impact on overall banking operations."
"Provide insights on customer segmentation, including average balances, transaction activity, and risk categories."

Here's a consolidated analysis across the three prompts you've been working with, based on the ICICI Bank customer dataset:

🚨 High-Risk Customers & Loan Exposure

- **Patterns Observed:**
  - Many **SME accounts** are flagged as **High Risk** with NPA status (e.g., *Priya Verma – CUST0007*, *Neha Patel – CUST0043*).
  - **Negative balances** and overdrafts are common in salary accounts (e.g., *Rahul Das – CUST0008*).
  - Several **closed accounts** still show high-risk flags, suggesting past defaults (*Varun Reddy – CUST0016*).
  - Some **active accounts** with high risk and ongoing loans (e.g., *Suresh Nair – CUST0010*) represent immediate exposure.
- **Loan Exposure:**
  - **SME loans** dominate the NPA list, highlighting vulnerability in small business lending.
  - **Corporate loans** show mixed risk—larger exposures but fewer defaults compared to SMEs.
  - **Retail salary accounts** often show overdrafts and negative balances, posing unsecured lending risks.

Insert Figure 5: Microsoft Copilot output for high-risk customer analysis and loan exposure insights.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

Summarize the key results and insights discovered from the project.

- **Data Quality Improvement:**

  After performing data cleaning, all inconsistencies, formatting errors, and missing values in the dataset were resolved. This ensured 100% usable records for analysis, improving the reliability of dashboard outputs.
- **High-Risk Customer Identification:**

  The AI-driven analysis identified high-risk customers, primarily concentrated in the SME and Retail segments, including accounts with NPAs, negative balances, and inactive accounts. This insight helps in proactive risk management.
- **Loan Exposure Insights:**

  The analysis revealed patterns of loan concentration and highlighted accounts with high exposure, NPAs, or past defaults. Segment-level trends indicated that SME loans carry higher systemic risk, while retail salary accounts often face overdraft issues.
- **AI Efficiency and Accuracy:**

  Microsoft Copilot significantly reduced manual analysis time. Tasks that previously took hours to analyze, such as identifying high-risk accounts and summarizing inactive accounts, were completed in minutes, with clear actionable recommendations.
- **Segmented Customer Insights:**

  Segmentation of customers into Retail, HNI, SME/Business, and Rural/Agri allowed for targeted insights:
- **Retail:** Inactive accounts, smaller loans, occasional defaults.
- **HNI:** High balances, diverse transactions, medium exposure risk.
- **SME:** High NPAs, frequent loan defaults, high-risk exposure.
- **Rural/Agri**: Seasonal activity, medium risk based on external factors.

- **Visualization Effectiveness:**

  The Power BI dashboard provided interactive visualizations of key metrics such as account status, loan exposure, risk category, and transaction trends. This enhanced decision-making by enabling quick interpretation of complex data.
- **Strategic Recommendations Derived:**

  Insights generated by AI enabled proactive decision-making, including early warning monitoring for SME and corporate accounts, auditing inactive accounts, and introducing targeted customer engagement and financial literacy programs.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project:Upload files · aditizasha-a11/ICICI-Bank-AI-Driven-Dashboard-
- Video Recording of Project Demonstration: https://drive.google.com/file/d/1UI3VAKhDUmiL_RsQ9ZhVJADy4a6cJjcW/view?usp=sharing

- Power BI / Dashboard Link (if applicable): [Paste published dashboard link here]

5.3 Limitations

Discuss the limitations of the current project or approach honestly.

- **Sample Dataset Constraints:**

  The project used a manually created synthetic dataset of 1,600 ICICI Bank customer records. While it simulates real-world data, the results may not fully generalize to larger or actual banking datasets with more diverse and complex customer behavior.
- **Limited AI Features:**

  Microsoft Copilot was the primary AI tool used. However, some advanced AI-assisted features were restricted due to access limitations, such as real-time predictive analytics or complex scenario simulations.
- **No Real-Time Data Integration:**

  The Power BI dashboard currently does not support real-time data refresh, meaning insights are based on static snapshots of the dataset. Dynamic, real-time monitoring would require additional infrastructure and live data feeds.
- **Scope of Risk Analysis:**

  Risk assessment focused mainly on high-level factors like NPAs, loan exposure, and account inactivity. Detailed behavioral or predictive credit scoring models were not implemented, limiting deeper predictive insights.
- **Visualization Scalability:**

  While the dashboard effectively visualizes the sample data, scaling to extremely large datasets or multiple branches may reduce performance unless optimized with advanced data modeling and aggregation.
- **Manual Verification Needed:**

  AI-generated insights and recommendations are helpful but not foolproof. Final decision-making still requires manual verification and domain expertise to avoid errors or misinterpretation.

5.4 Future Work

Provide suggestions for improving or extending the project in the future.

- **Real-Time Data Integration:**

  Implement live data pipelines using tools like Power Automate, API connections, or Azure Data Services, enabling the dashboard to reflect real-time customer activity and financial health.

- **Advanced Analytics with Machine Learning:**
  Apply predictive and prescriptive models, such as credit risk scoring, loan default forecasting, and anomaly detection, on the cleaned dataset to enhance proactive decision-making.
- **Web-Based Dashboard / User Interface:**
  Develop a web application or Power Apps interface for non-technical users, allowing interactive exploration of data insights, filtering options, and AI-assisted recommendations.
- **Expanded and Diverse Dataset:**
  Include larger and more varied datasets, incorporating multiple branches, different account types, and historical financial transactions to improve the generalizability and accuracy of insights.
- **AI Prompt Library Development:**
  Create a standardized library of prompts for Microsoft Copilot to quickly generate insights, reports, and summaries, making AI-assisted analytics more consistent and efficient.
- **Enhanced Risk & Compliance Monitoring:**
  Incorporate modules for regulatory compliance checks, fraud detection, and portfolio stress testing to provide a more comprehensive financial risk management tool.
- **Integration with Business Processes:**
  Connect dashboard insights with internal banking workflows, such as targeted customer campaigns, loan restructuring strategies, and credit limit adjustments, to make analytics directly actionable.

5.5 Conclusion

This project successfully demonstrated how Microsoft Excel, Power BI, and Microsoft Copilot can be leveraged to clean, analyze, and visualize complex banking datasets. The primary objective was to transform raw ICICI Bank customer data into actionable insights, and this was achieved through a structured methodology involving data cleaning, analysis, visualization, and AI-assisted recommendations.

Key findings include the identification of high-risk customers, loan exposure patterns, and inactive accounts, along with segment-level insights that help tailor banking strategies for Retail, SME, Corporate, and Salary account customers. The interactive dashboard provides a centralized view of financial health and risk levels, supporting faster and more informed decision-making.
All project objectives were met, including data standardization, KPI calculation, AI-powered insight generation, and visualization of trends. The use of Microsoft Copilot significantly enhanced efficiency, reduced manual effort, and provided structured insights that would otherwise take hours to compile.

Personally, this fellowship experience strengthened my practical skills in data analytics, AI-assisted reporting, and dashboard creation, while also providing exposure to real-world financial analysis challenges. The methodology and tools explored in this project can be extended to other domains such as healthcare, retail, or investment analytics, making it a scalable and valuable framework for future projects.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

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[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

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APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

Include a sample of the raw dataset used in the project. This may consist of the first 10–20 rows of data or a summary of key fields such as customer ID, age, account balance, transaction details, and loan information.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Customer	Customer_Name (St	Customer	Account_	Branch_R	Account_	Opening_	Total_Cre	Total_Del	Average_	Current_E	Loan_Am	Loan_Stat	Account_	Risk_Cate	Transactio
2	CUST0001	Vivaan Kapoor	SME	Current	West	22/08/202	179093.1	294928	203891.6	224611.3	270129.5	152681.4	Closed	Inactive	Medium	176
3	CUST0002	Ishita Mehta	SME	Savings	North	02/01/202	191192.5	156187	121939	208316.5	225440.5	238716.9	NPA	Inactive	Medium	17
4	CUST0003	Manish Yadav	SME	Savings	South	06/09/202	42866.88	401588.3	50395.26	218463.4	394059.9	253163.5	NPA	Inactive	Low	80
5	CUST0004	Akash Malhotra	Corporate	Savings	North	19/04/201	47920.74	324981.5	248119.2	86351.91	124783.1	157731	NPA	Inactive	Low	132
6	CUST0005	Manish Gupta	Retail	Business	East	01/03/202	161982.6	163289.7	264376	111439.5	60896.29	118309	NPA	Inactive	Low	177
7	CUST0006	Akash Reddy	SME	Business	North	22/01/202	37137.13	137605.4	153313.9	29282.88	21428.63	60852.51	Closed	Active	Medium	117
8	CUST0007	Priya Verma	SME	Savings	West	18/02/201	94376.26	157636.9	99379.5	123505	152633.7	248598.9	NPA	Inactive	High	104
9	CUST0008	Rahul Das	SME	Salary	North	29/12/202	51015.64	35846.18	393774.3	-127948	-306912	57853.99	Closed	Active	High	6
10	CUST0009	Aman Bansal	Corporate	Current	East	06/07/201	190146.1	32045.27	207674	102331.7	14517.34	25086.95	No Loan	Inactive	Medium	133
11	CUST0010	Suresh Nair	SME	Salary	North	06/02/202	183912.8	79103.12	95043.84	175942.5	167972.1	80206.54	Active	Active	High	187
12	CUST0011	Shreya Verma	Retail	Savings	South	04/10/201	75111.16	157910.3	50818.52	128657.1	182202.9	212791.5	Active	Inactive	High	161
13	CUST0012	Varun Reddy	SME	Salary	South	06/11/201	167089.5	421129.4	362321.9	196493.3	225897	114500.9	NPA	Inactive	Low	35
14	CUST0013	Varun Nair	Retail	Salary	West	13/08/202	9944.47	167633	290385	-51431.5	-112807	40058.92	No Loan	Active	Low	126
15	CUST0014	Aditi Singh	Corporate	Current	East	21/04/201	89656.24	163464	185148.2	78814.11	67971.97	215329.9	No Loan	Active	Medium	38
16	CUST0015	Simran Kapoor	Corporate	Business	North	22/11/202	190382.2	418267.7	354817.9	222107.2	253832.1	168545.7	Active	Inactive	Medium	200
17	CUST0016	Varun Reddy	SME	Business	South	24/07/201	163440.7	145128.1	208495.2	131757.2	100073.6	243471.5	Closed	Inactive	High	192
18	CUST0017	Diya Gupta	Corporate	Salary	South	05/04/201	107775	41363.5	363377.5	-53231.9	-214239	144401.2	Active	Inactive	Low	84
19	CUST0018	Manish Patel	SME	Business	North	20/10/201	140199.8	306666.6	106390	240338.1	340476.4	236600.1	No Loan	Active	Medium	66
20	CUST0019	Shreya Verma	SME	Business	North	26/07/202	23465.74	376230	299229.2	61966.15	100466.6	240555.4	Active	Active	Medium	136
21	CUST0020	Simran Patel	Retail	Salary	South	23/04/202	128273.7	14166.66	154935.2	57889.41	-12494.9	233292.4	No Loan	Active	Low	142
22	CUST0021	Suresh Khan	SME	Business	North	27/08/202	42752.07	222365.9	346412.4	-19271.2	-81294.4	181084.2	Active	Inactive	Low	182
23	CUST0022	Vihaan Verma	Corporate	Salary	North	22/08/201	104670.2	52833.85	94987.36	83593.48	62516.72	227026.5	Closed	Active	Low	191
24	CUST0023	Aditya Nair	Retail	Salary	South	11/07/201	12998.25	200466.8	274989.5	-24263.1	-61524.4	134643.6	NPA	Inactive	Low	154
25	CUST0024	Vivaan Reddy	SME	Current	West	08/12/202	158501.5	452852.3	204048.7	282903.2	407305	84630.13	No Loan	Active	Medium	127
26	CUST0025	Vihaan Joshi	Retail	Salary	West	25/05/202	158149.9	343103.2	14331.65	322535.6	486921.4	79248.27	NPA	Inactive	Low	56
27	CUST0026	Ananya Singh	SME	Salary	North	04/05/201	98931.68	11434.92	137296.2	36001.05	-26929.6	164076.7	Active	Inactive	Low	41
28	CUST0027	Diya Mehta	Retail	Salary	West	13/11/201	5938.22	174323.2	306492.6	-60146.5	-126231	140314.3	No Loan	Active	High	24
29	CUST0028	Shreya Yadav	Retail	Current	South	14/07/201	31689.4	20207.69	270643	-93528.2	-218746	117231.5	Active	Inactive	Low	42
30	CUST0029	Aditya Mehta	Retail	Salary	South	07/08/202	97849.51	93238.51	174261.4	57338.09	16826.67	201673.1	Active	Inactive	Low	59
31	CUST0030	Nikhil Reddy	SME	Business	South	06/01/201	188116.6	124200.4	105176	197628.9	207141.1	152414.1	Active	Inactive	Low	90
32	CUST0031	Sneha Khan	Corporate	Salary	North	20/01/201	91311.67	323438	275891.8	115084.8	138857.9	6326.96	Closed	Active	Low	100
33	CUST0032	Shreya Kapoor	SME	Current	West	08/09/201	71851.31	263539	353416.2	26912.74	-18025.8	139478.9	No Loan	Inactive	Low	74
34	CUST0033	Sai Patel	SME	Salary	West	08/08/202	112692.6	32119.65	61165.15	98169.83	83647.08	225323.2	No Loan	Inactive	Medium	80
35	CUST0034	Sneha Das	Retail	Savings	East	15/11/202	125413.7	91774	349644.7	-3521.67	-132457	58853.62	Active	Active	Medium	65
36	CUST0035	Kavya Joshi	Corporate	Salary	South	09/08/202	111164.4	302461	283691.5	120549.2	129933.9	3181.56	No Loan	Active	Low	39
37	CUST0036	Meera Malhotra	SME	Current	West	30/07/202	52749.75	22503.82	39716.76	44143.28	35536.81	107159	No Loan	Inactive	High	12
38	CUST0037	Meera Verma	SME	Savings	North	02/10/201	112736.5	51104.45	277329.5	-376.03	-113489	32272.75	No Loan	Inactive	Medium	108
39	CUST0038	Kavya Kapoor	Retail	Current	South	13/09/201	63631.09	497523	166969.5	228907.8	394184.6	25591.76	No Loan	Active	High	97
40	CUST0039	Vivaan Khan	SME	Current	South	22/07/202	120151.3	66402.69	75612.7	115546.3	110941.3	10555.27	Closed	Active	Low	7

Insert Table 1: Sample of Raw ICICI Bank Dataset (Before Data Cleaning)

Description:

The above dataset represents a sample of the raw customer data used in this project. It includes key variables such as customer demographics, account details, transaction records, and loan-related information. The dataset initially contained inconsistencies such as missing values, duplicate

entries, and formatting issues. These issues were identified and resolved during the data cleaning and preprocessing stage to ensure accuracy, reliability, and readiness for further analysis.

Appendix B — Key Excel Formulas / DAX Measures Used

The following formulas and measures were used for data cleaning, validation, and analysis:

- =IFERROR(VLOOKUP(A2, Sheet2!A:B, 2, FALSE), "Not Found")
  Used for validating and retrieving missing data
- =COUNTIF(A:A, A2)
  Used to identify duplicate records
- =IF(B2="", "Missing", B2)
  Used to handle missing values
- Power BI DAX:
  Total Balance = SUM(CustomerData[Account Balance])
  Used to calculate total account balance
- High Risk Customers = CALCULATE(COUNT(CustomerData[Customer ID]), CustomerData[Risk Level] = "High")
  Used to identify high-risk customers

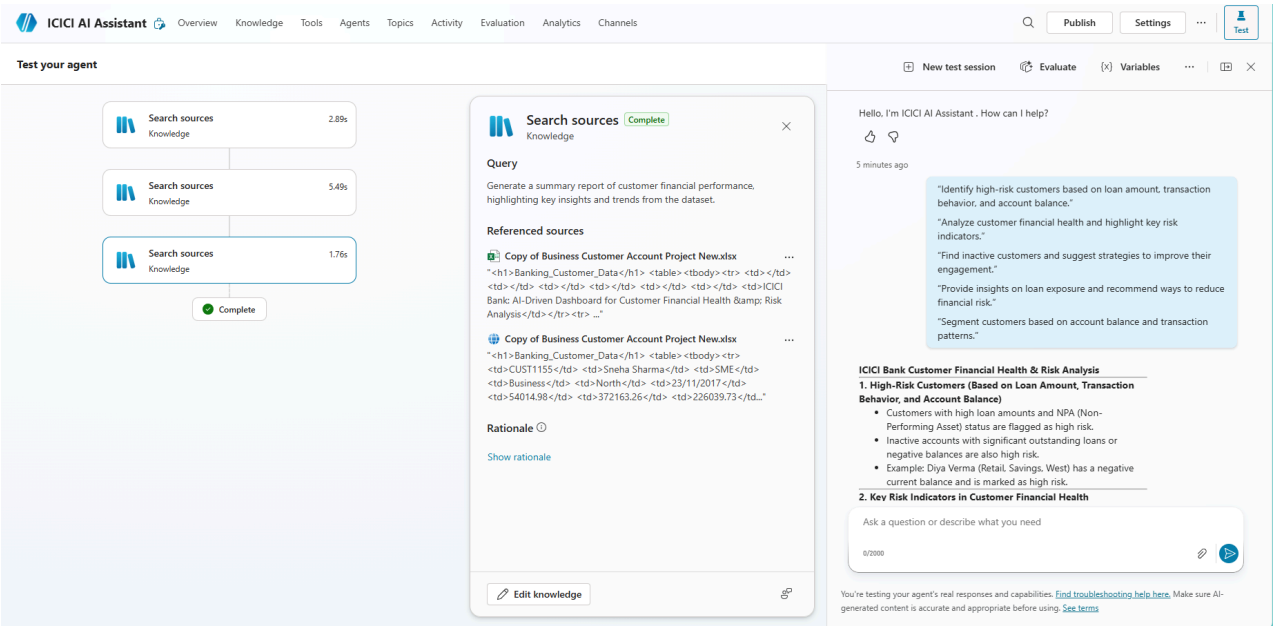
Appendix C — AI Prompts Used

The following prompts were used with Microsoft Copilot and ICICI AI Assistant:

- Prompt 1:
  “Analyze the customer dataset and identify high-risk customers based on loan amount, transaction patterns, and account balance.”
- Prompt 2:
  “Generate a summary of customer financial health and highlight key risk indicators.”
- Prompt 3:
  “Identify inactive accounts and suggest possible actions to improve customer engagement.”
- Prompt 4:
  “Provide insights on loan exposure and recommend strategies to reduce financial risk.”

Appendix D — Screenshot Gallery

This section includes additional screenshots and visual outputs generated during the project that were not included in the main chapters. These visuals provide deeper insights into customer segmentation, financial risk analysis, and account activity patterns, along with AI-assisted outputs generated using the ICICI AI Assistant.

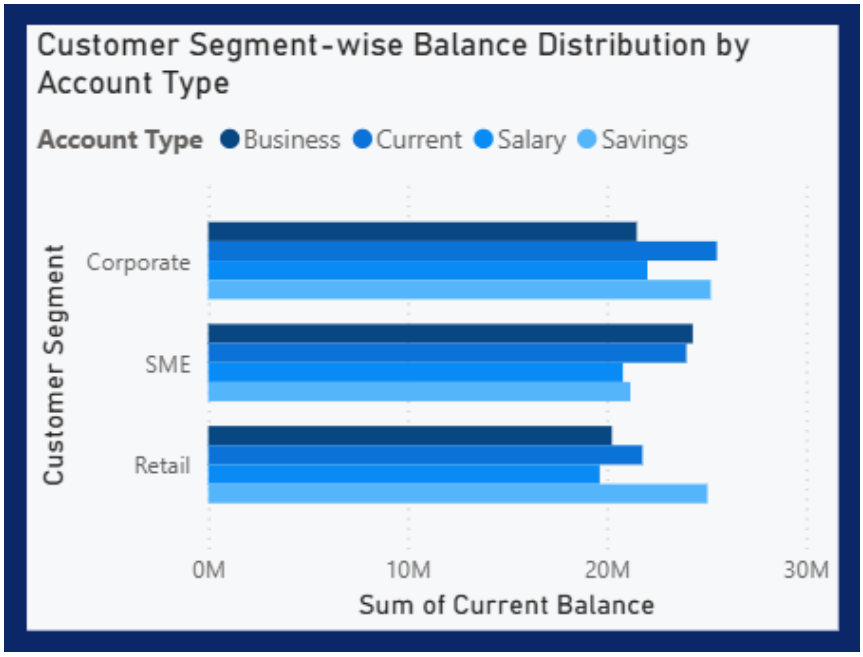


Insert Figure 2: ICICI AI Assistant Output for Customer Risk Analysis

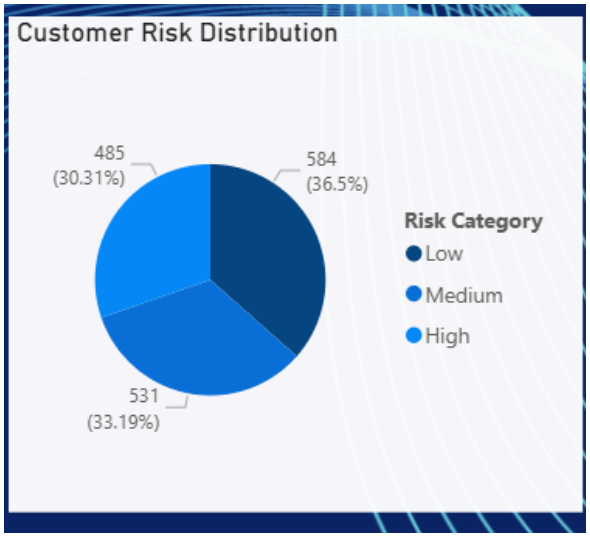
Sample Queries Used with ICICI AI Assistant

- **"Identify high-risk customers based on loan amount, transaction behavior, and account balance."**
- **"Analyze customer financial health and highlight key risk indicators."**
- **"Find inactive customers and suggest strategies to improve their engagement."**
- **"Provide insights on loan exposure and recommend ways to reduce financial risk."**
- **"Segment customers based on account balance and transaction patterns."**

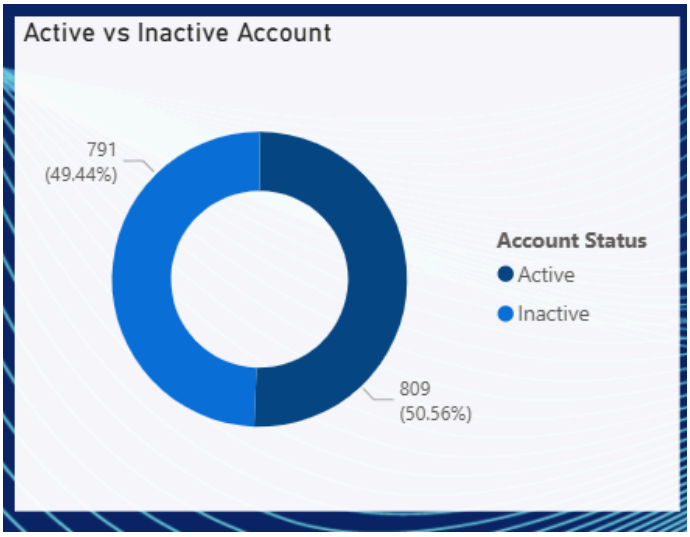
Description: The above queries were used to generate AI-driven insights and automate financial analysis using the ICICI AI Assistant.



Insert Figure: Customer Segment-wise Balance Distribution



Insert Figure 3: Customer Risk Distribution Analysis



Insert Figure 4: Active vs Inactive Customer Analysis